MGS 602: IT & Cloud Infrastructure Management

Introduction to AWS IAM

1) Imagine you are an administrator managing the IAM users and groups for a large organization. What are some challenges you might face in assigning and managing permissions for a large number of users and groups?

Response: There can be difficulties like-

As the organization expands, it becomes more challenging to organize and manage permissions for a growing number of users and groups. Monitoring Adjustments can be an issue, more precisely, maintaining record of individuals with permissions and ensuring security and compliance can present a significant challenge. Also, with teams evolving and expanding, it can be challenging to ensure groups are kept current and users are correctly assigned. When it comes to that, finding the appropriate level of access for various teams can be challenging in order to avoid granting excessive or insufficient permissions.

Last but not the least, users may often acquire an excess of permissions beyond their necessity, resulting in potential security hazards.

2) Think about a real-world scenario in which you might need to assign permissions to users and groups in AWS IAM. Describe the process you would follow to ensure that each user has the appropriate level of access to AWS resources while maintaining security.

Response: Here is the procedure I will follow to grant permissions using AWS IAM while still maintaining security.

Initially, I would determine the tasks required for each team to complete. One scenario could involve developers requiring access to EC2 and repositories for code, while analysts require access to databases and data stored in S3. Followed by establishing User Groups i.e. classifying users according to their roles. For example, form categories for "Programmers," "Data Specialists," and "Customer Service Representatives." This makes managing permissions easier by assigning policies to groups rather than to individuals. In addition to this, I will develop policies for each group that provide them with the least amount of permissions required to complete their responsibilities. Developers may have the privilege to control the start/stop function of EC2 instances without having access to billing details.

Next, establishing IAM policies for every group. Developers may have an EC2 policy allowing them to manage instances, whereas analysts might be limited to a read-only policy for S3. Connecting these guidelines to the appropriate groups. In this manner, users are able to take on roles as required, minimizing unneeded access. Multi-factor authentication could be used in order to add additional level of security.

3) How do IAM policies work in AWS IAM, and what are the effects of policies on service access? Can you give an example of a policy and explain its basic structure?

Response: In AWS IAM, policies are JSON documents that control what actions users, groups, or roles can perform on AWS resources. Each policy specifies the operations allowed or denied (e.g., ec2:StartInstances), AWS resources the actions apply to (e.g., EC2 instances in a specific region) and whether the action is Allow or Deny.

Example of a policy to start and stop EC2 instances in the **us-west-2** region:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "arn:aws:ec2:us-west-2:123456789012:instance/*"
    }
  ]
}
```

Impact:

The effect is that only the specified actions are permitted ("Allow"). The user is allowed to start (ec2:StartInstances) and stop (ec2:StopInstances) EC2 instances. And the policy applies to all EC2 instances in the us-west-2 region under the account 123456789012.

With this policy, the user can only start and stop EC2 instances in the specified region, but cannot terminate or modify them without additional permissions.

4) In this lab, you were given pre-created IAM users and groups. What are some ways in which you could create your own IAM users and groups, and how might you manage their permissions?

Response: I can create IAM users and groups using:

1. AWS Management Console: by creating users in the "Users" section and groups in the "Groups" section. Attach policies to manage permissions.

2. AWS CLI: by using commands like `aws iam create-user` for users and `aws iam create-group` for groups. Manage permissions with `aws iam attach-user-policy` or `aws iam attach-group-policy`.

3. AWS SDKs: by automating user and group creation with programming languages using relevant SDK functions.

4. Infrastructure as Code (IaC): by using AWS CloudFormation or Terraform to define IAM resources in code and manage permissions through IAM policies.

<u>Managing Permissions</u>-

a) IAM Policies: defining granular permissions for actions on resources.

b) Managed Policies: using AWS-managed or customer-managed policies.

c) Group Policies: assigning permissions at the group level for all users in that group.

d) Permission Boundaries: by controlling maximum permissions for IAM users.

5) What is the difference between a managed policy and an inline policy in AWS IAM, and in what situations might you choose to use one over the other?

Response: In AWS IAM, a managed policy is an independent policy that can be applied to various users, groups, or roles, created by either AWS or the user. It's wonderful to have uniform permissions for numerous identities. On the other hand, an inline policy is directly linked to a specific user, group, or role and is specific to that entity.

I would prefer inline policies for more personalized or short-term access needs whereas I would use managed policies for reusability when I would want to use it in longer-term,

6) What is the difference between a user and a role in AWS IAM?

Response: In AWS IAM, a **User** is an individual with long-term credentials, while a **Role** can be temporarily assumed by trusted entities and uses temporary credentials. Users have enduring access, while roles provide temporary access or permissions across multiple accounts.

7) Can you explain the concept of least privilege and how it relates to AWS IAM?

Response: The idea of least privilege involves giving users and roles only the essential permissions for their job tasks, reducing the chance of unintentional or harmful actions. In AWS IAM, this concept is implemented through the creation of policies that permit only the essential actions on the required resources, preventing users from having excessive access. This increases safety by decreasing the possible areas for attack.

8) How can you use IAM groups to manage permissions for multiple users, and what are some best practices for using IAM groups?

Response: Using IAM groups in AWS effectively manages permissions for numerous users. Creating groups based on roles or job functions allows you to grant permissions to the group as a whole instead of granting them individually to each user. This simplifies access management by allowing for easy updates or removals from a group when a user changes roles or leaves the organization instead of adjusting permissions for each user separately.

The suggested methods for utilizing IAM groups are-

1. Group by Job Title: Form teams according to similar roles (such as developers, administrators, analysts) for easier management of permissions.

2. Restrict Group Size: Maintain small, specialized groups to prevent wide-ranging access and simplify auditing processes.

4. Frequently Monitor Group Membership: Regularly review group memberships and permissions to make sure they match current roles and duties.

5. Enforce Least Privilege: Make sure that groups are only granted the permissions they need for their designated duties, reducing potential security threats.

By adhering to these methods, you can enhance access control and increase the security of your AWS setup.

9) Provide a brief executive summary of the lab you completed along with screenshots.

Response: In this lab activity, I worked upon AWS Lab and examined Identity and Access Management (IAM) to administer access rights according to job positions.

First of all, three users; **user-1, user-2, and user-3**, were created with no permissions. Post which, three groups were formed: **EC2-Admin, EC2-Support, and S3-Support**. Followed by that, assignments were given to the users in a way-

a. user-1: Included in S3-Support is the ability to view S3 with read-only permission.

b. user-2: EC2-Support was included for read-only entry to EC2.

c. user-3: EC2-Admin was given permission to manage EC2 instances.

By effectively adhering the respective services to the respective users, signing in and checking whether the users are added in the respective service, and logging out, I managed to effectively complete the first lab and achieved the score of 100%.

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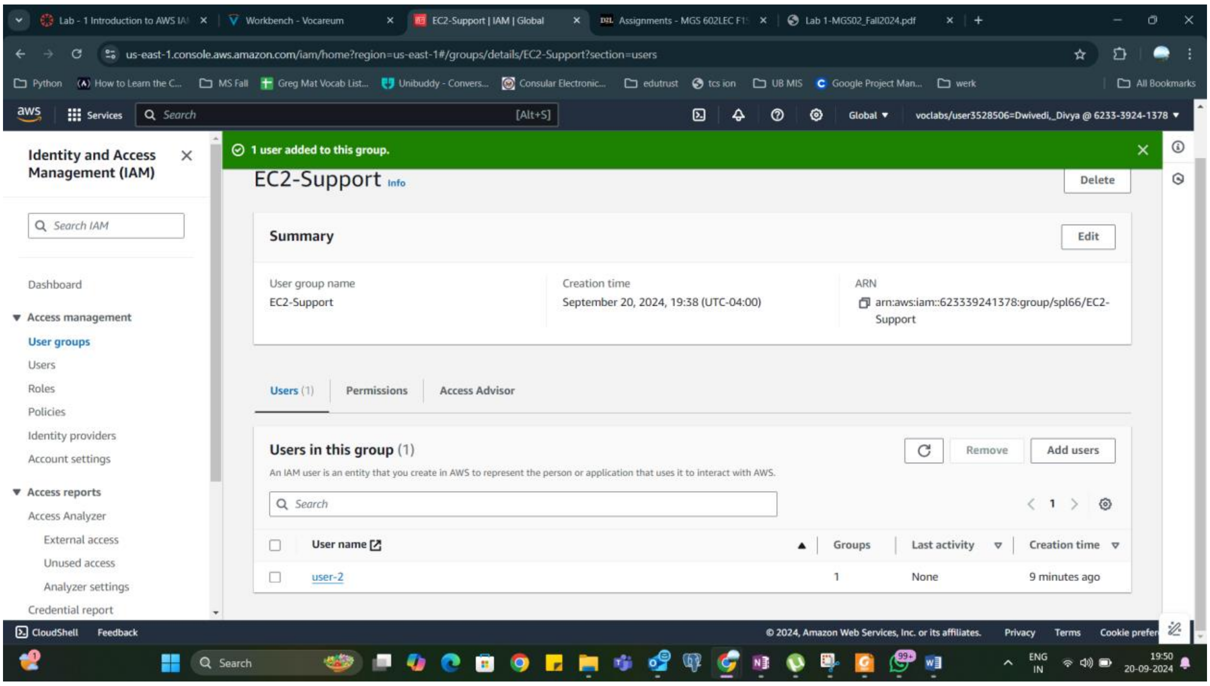
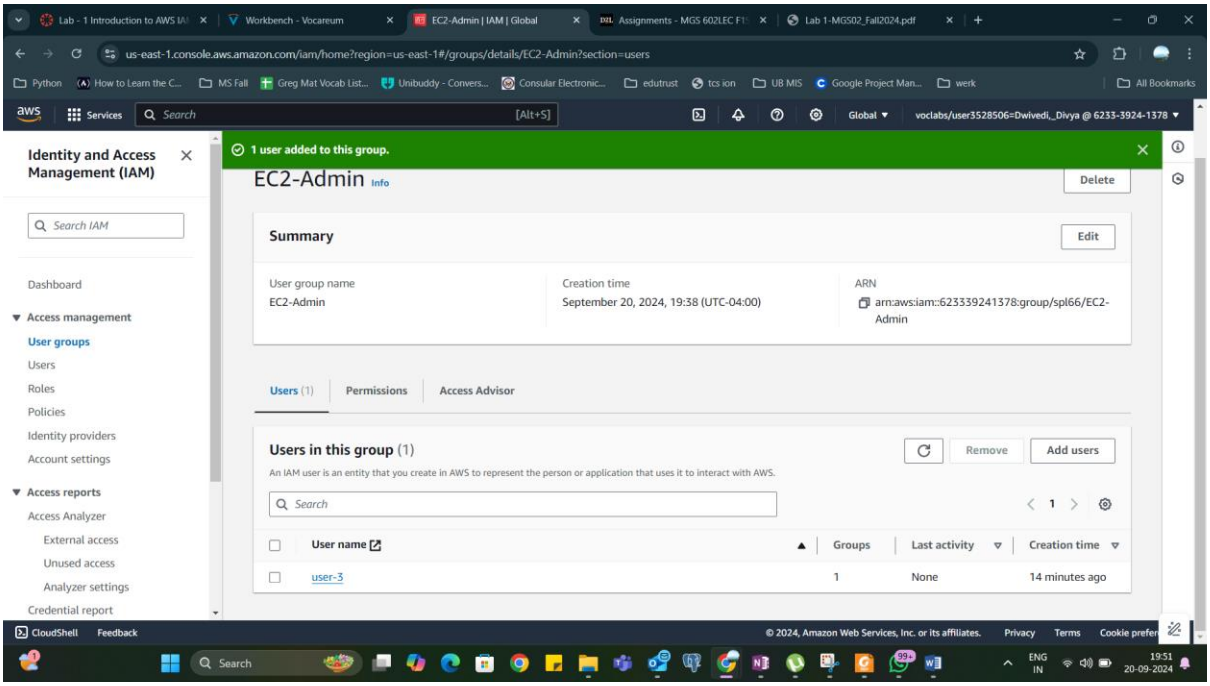
SCREENSHOTS-





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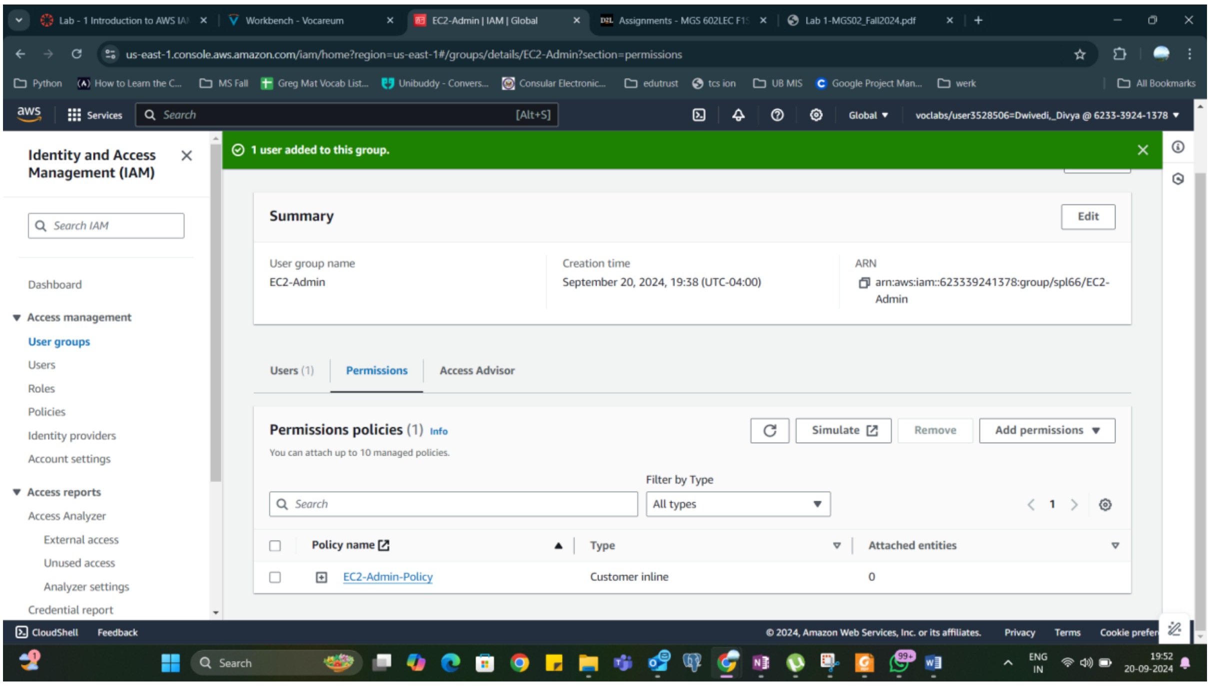

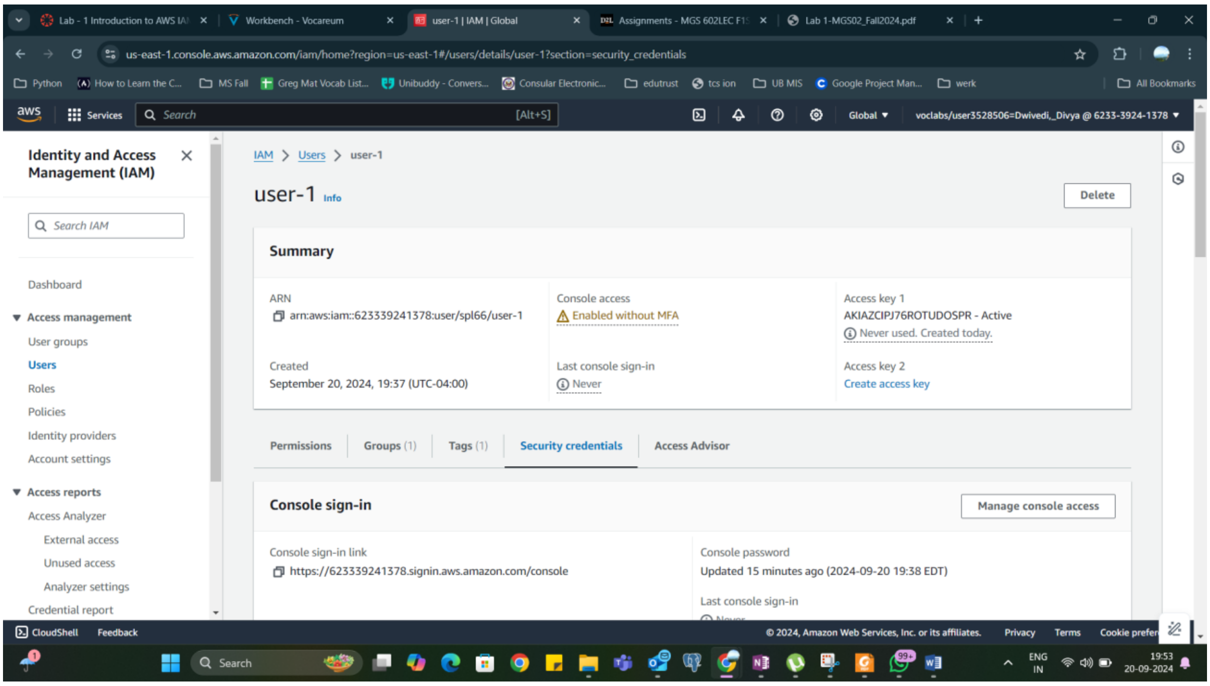

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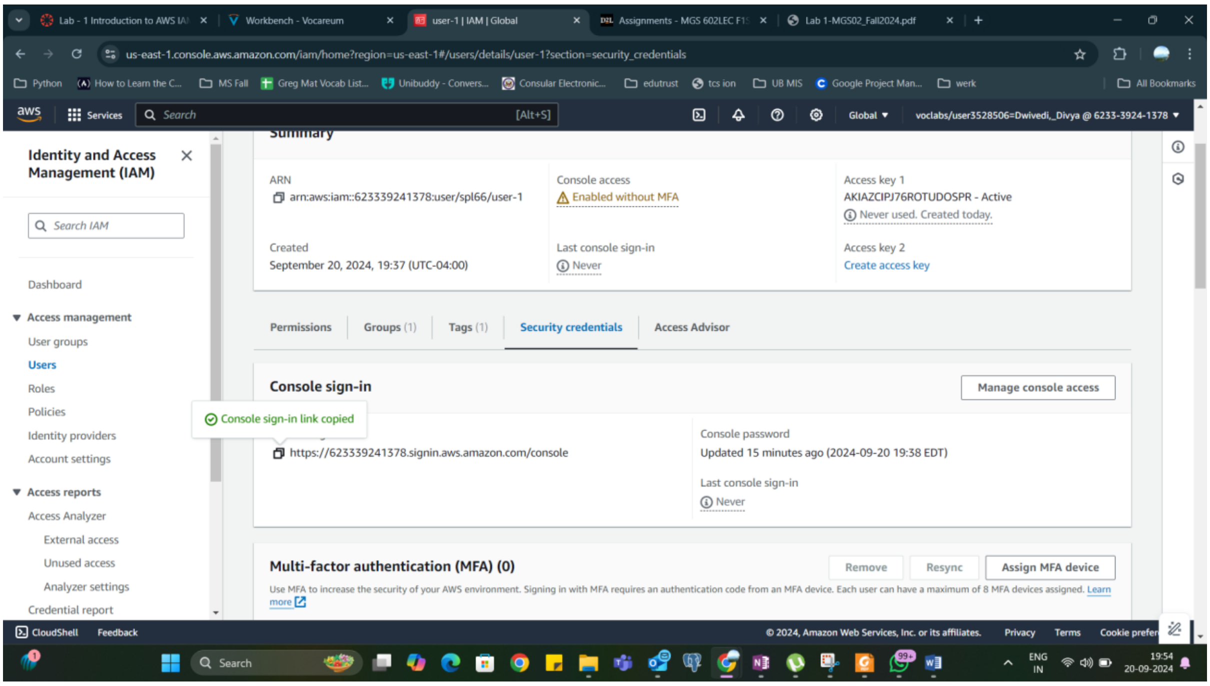

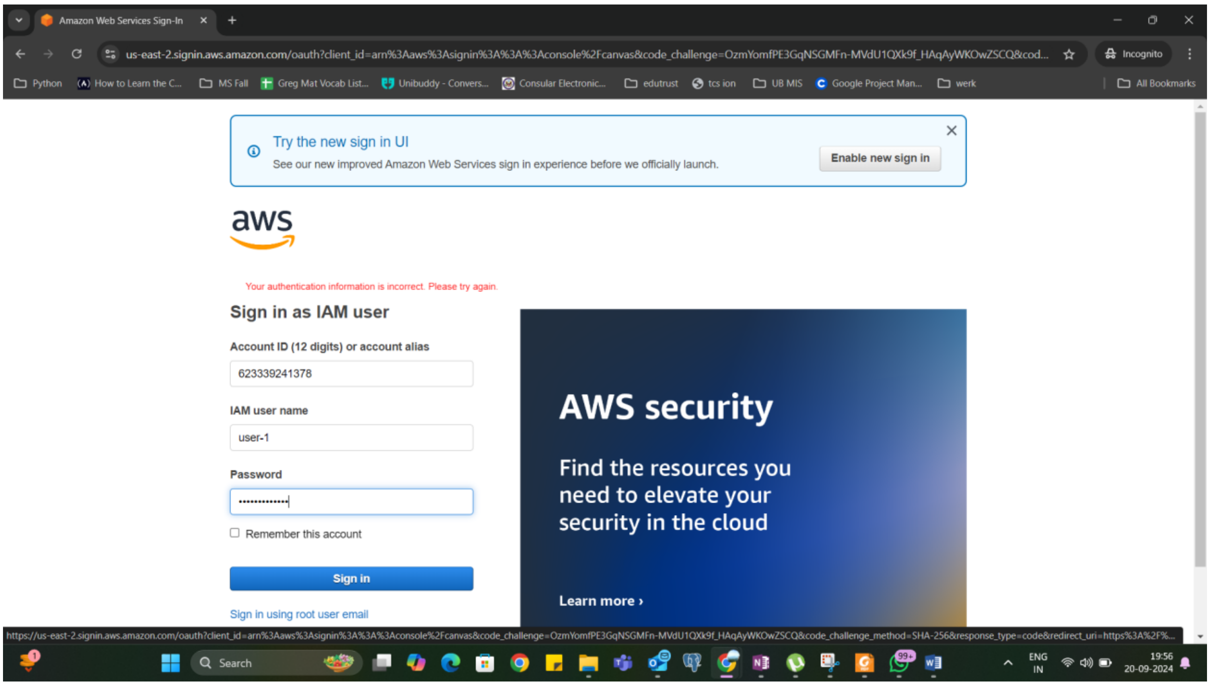

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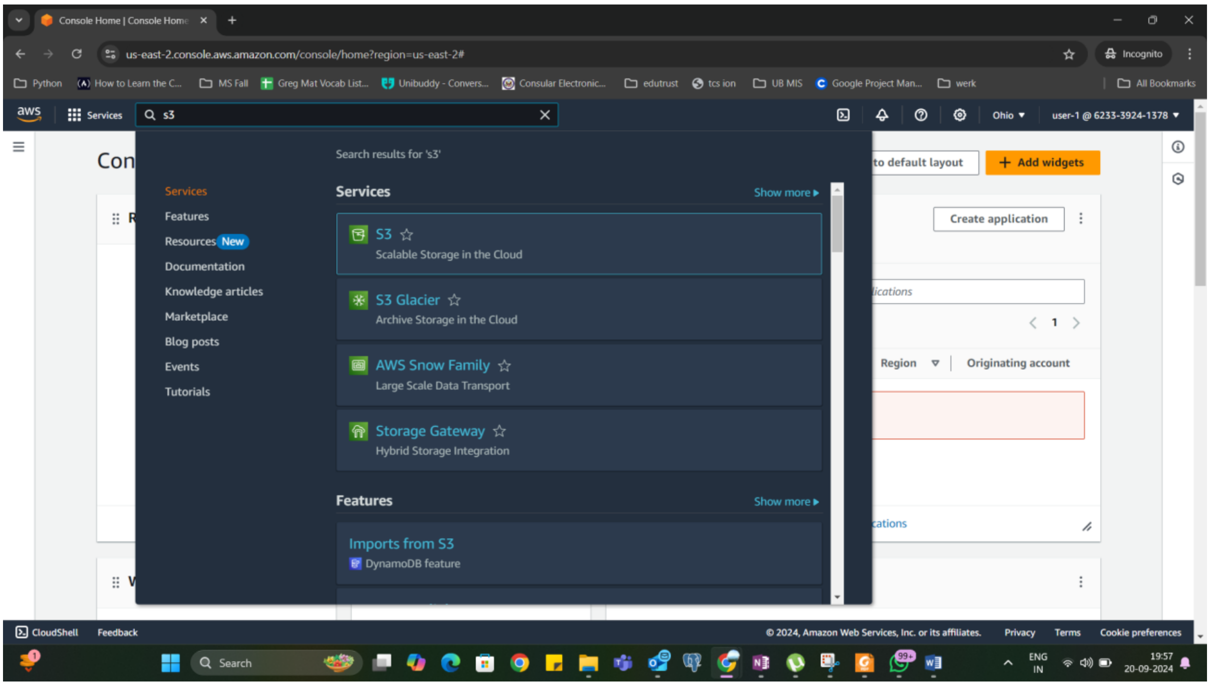

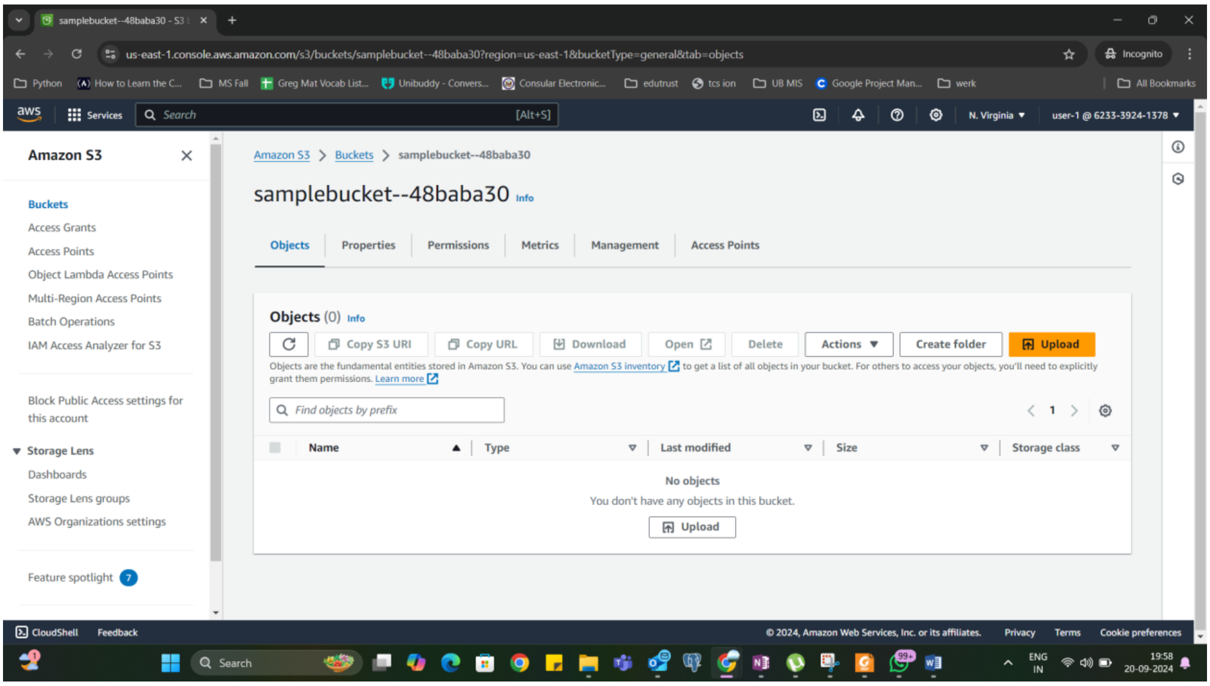

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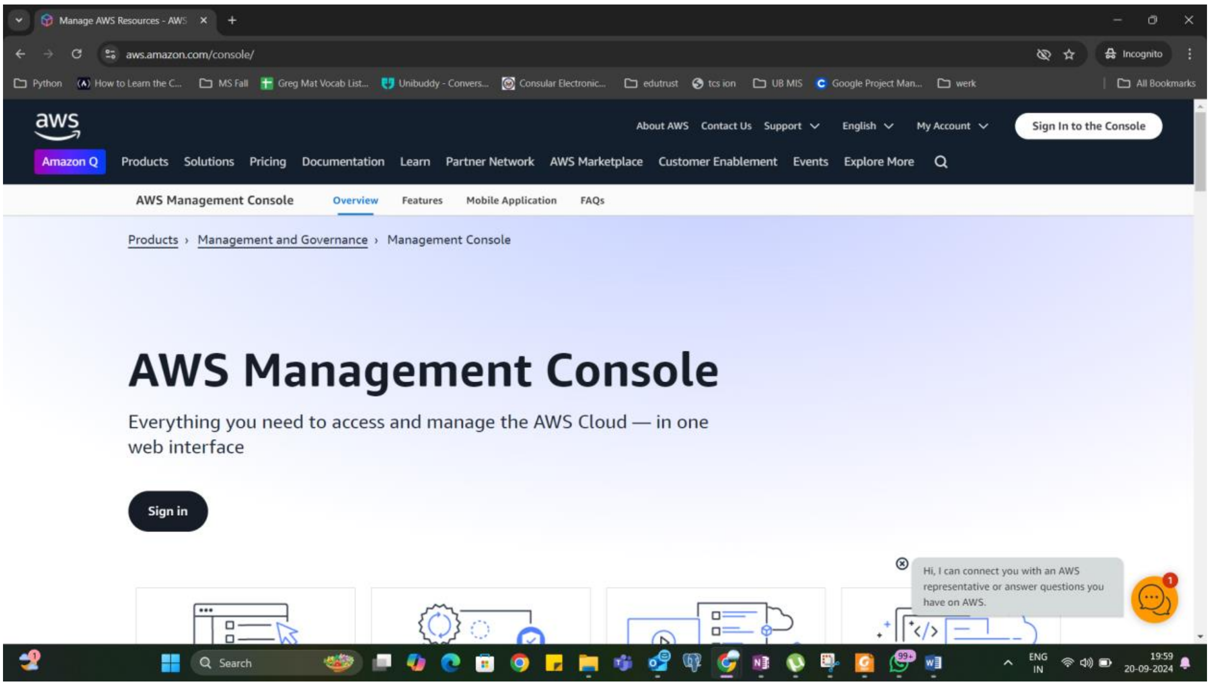

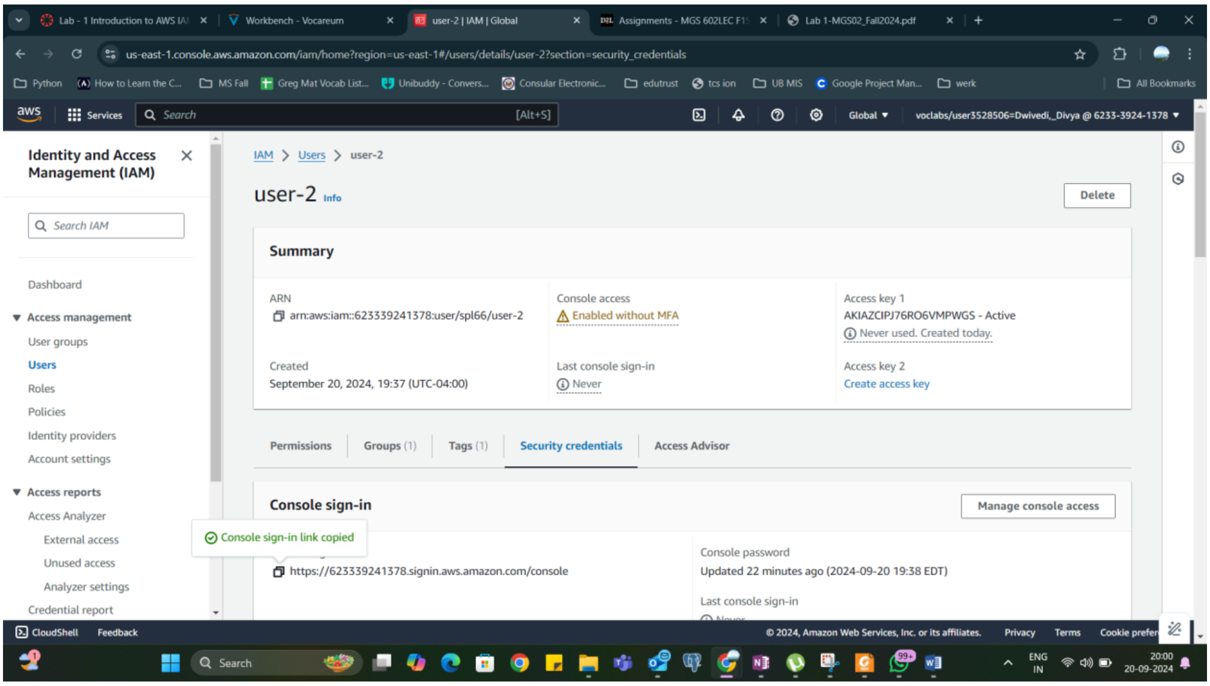

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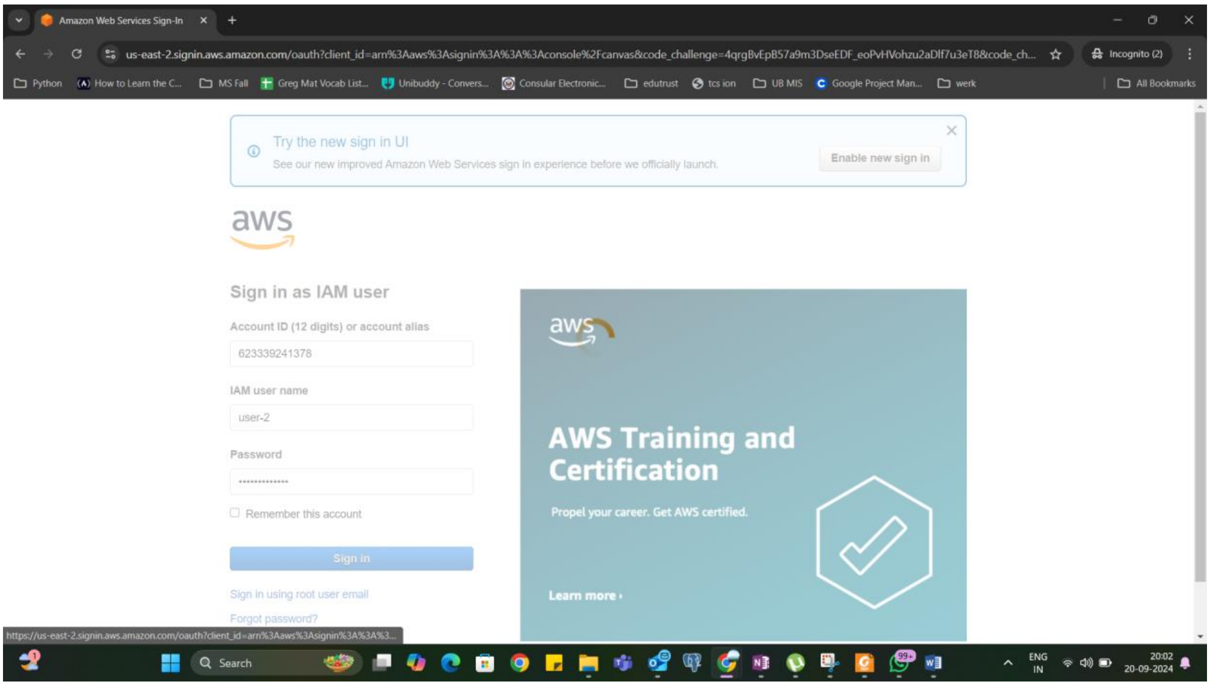

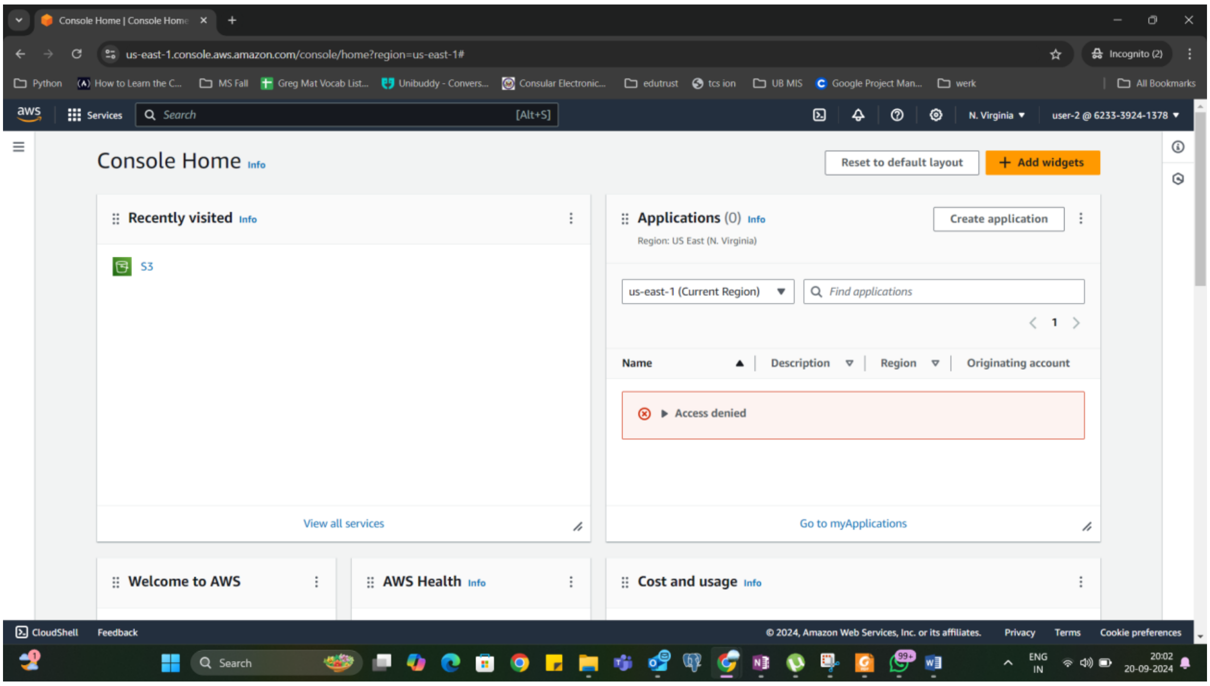

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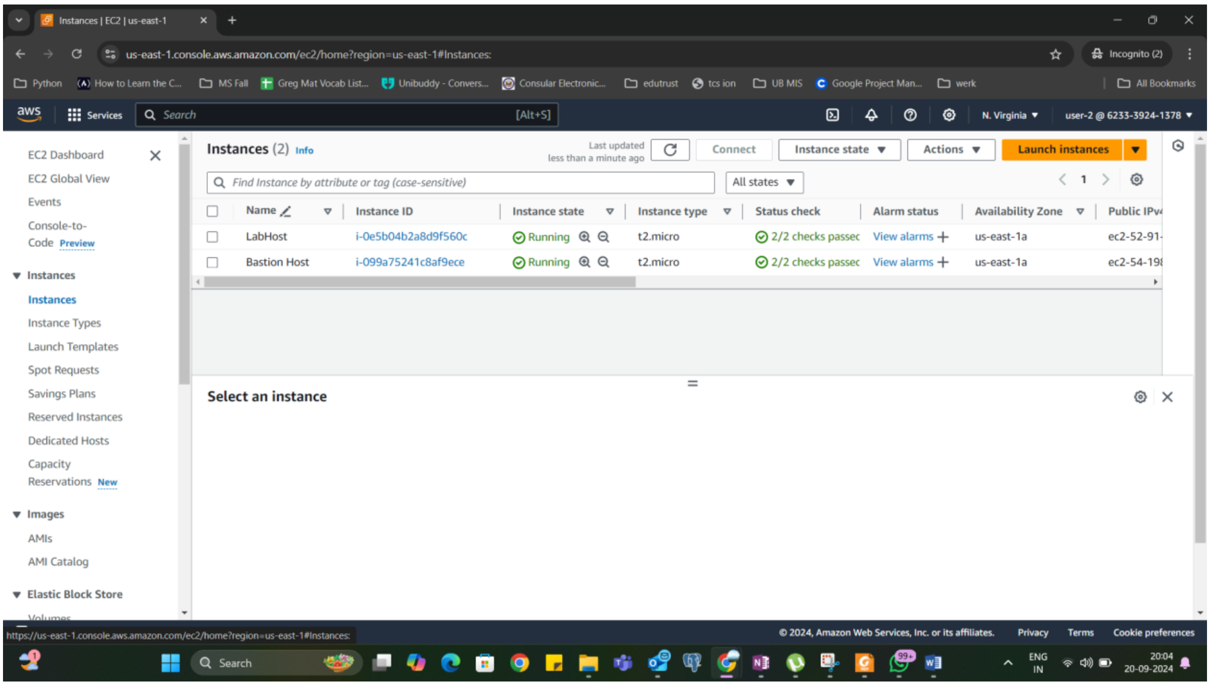

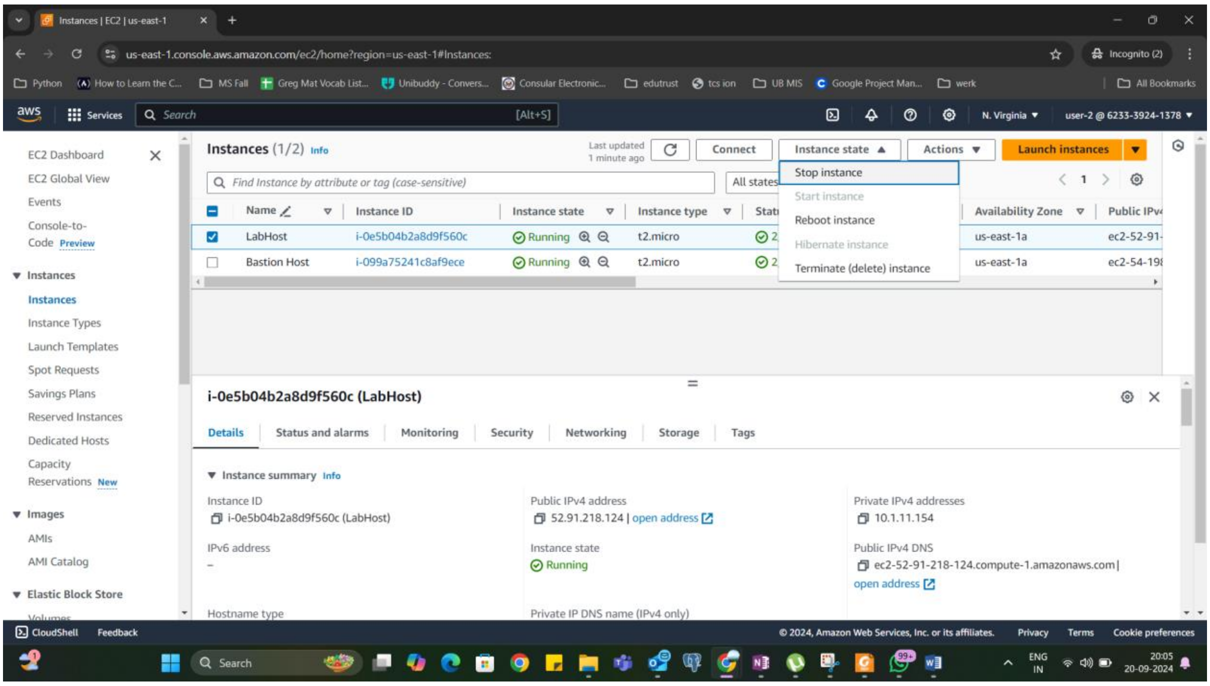

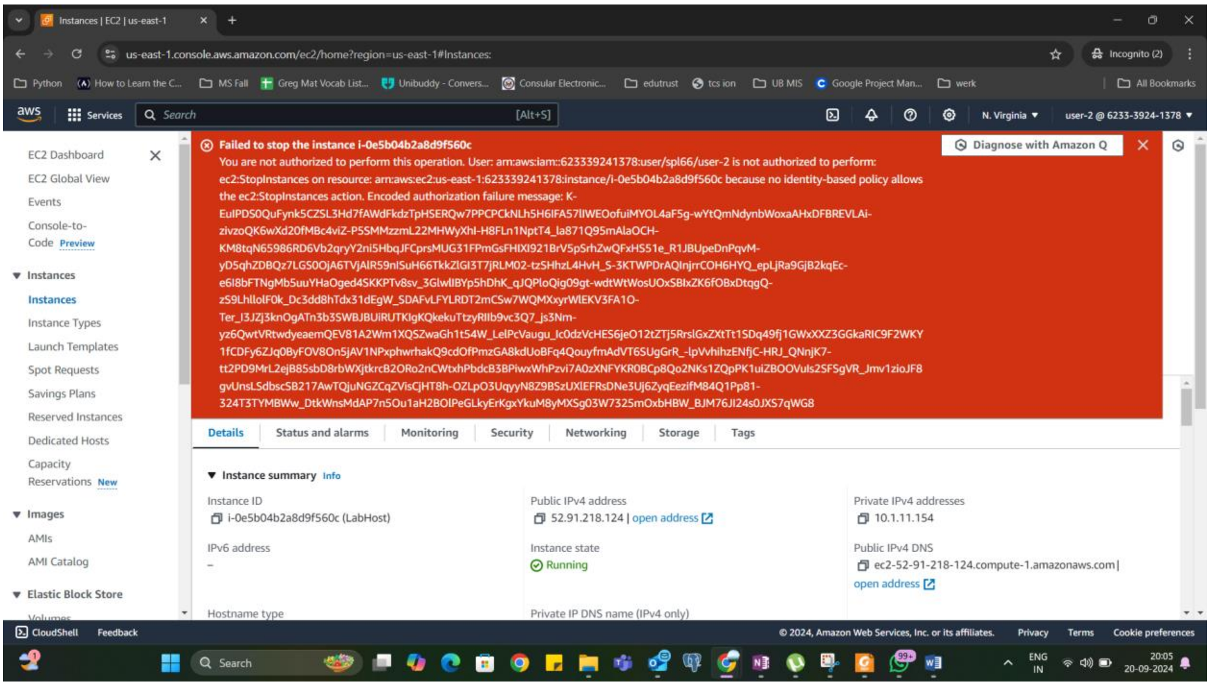

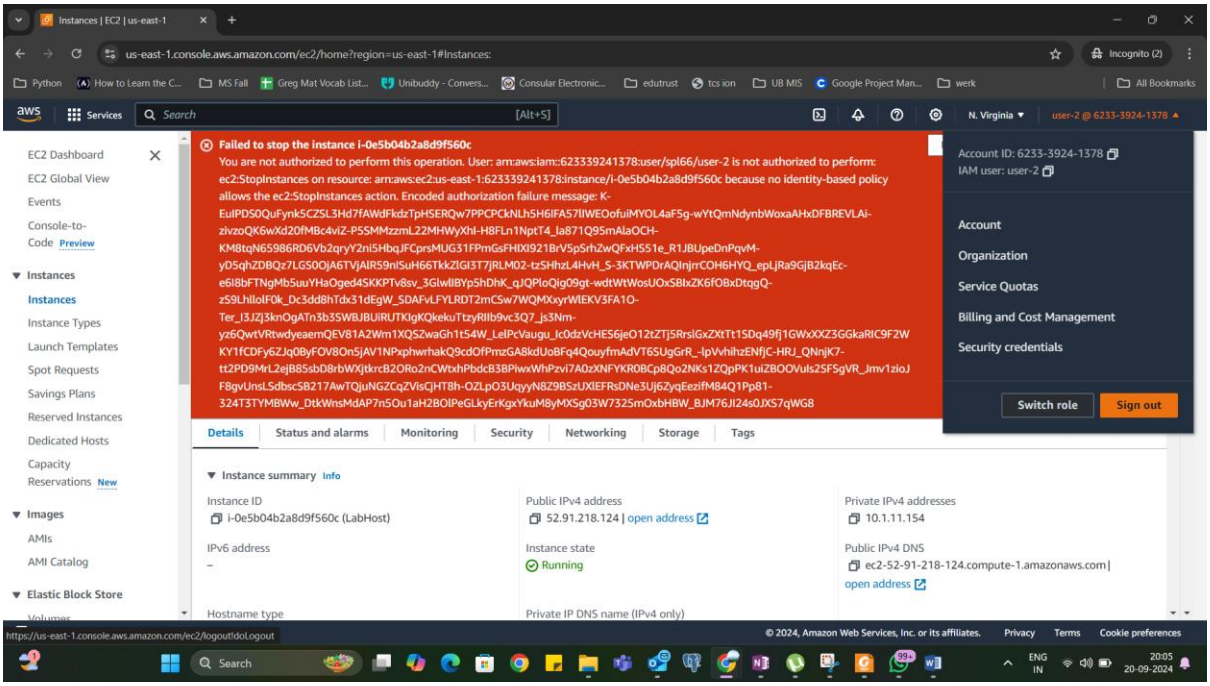

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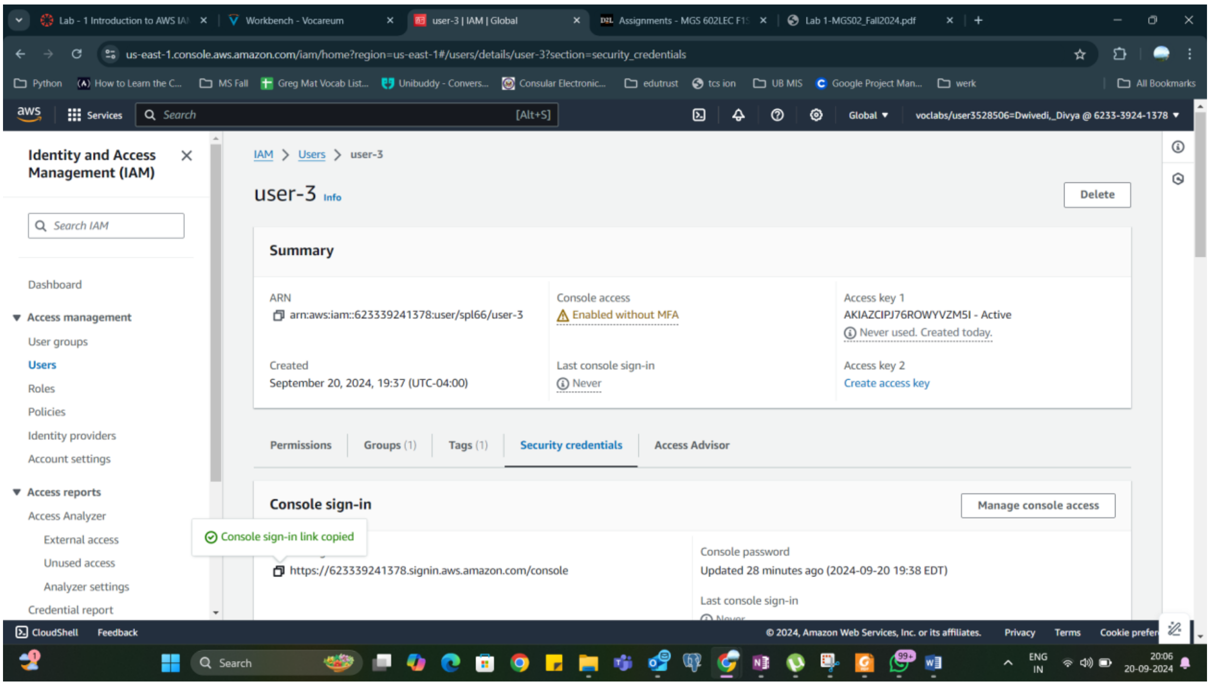

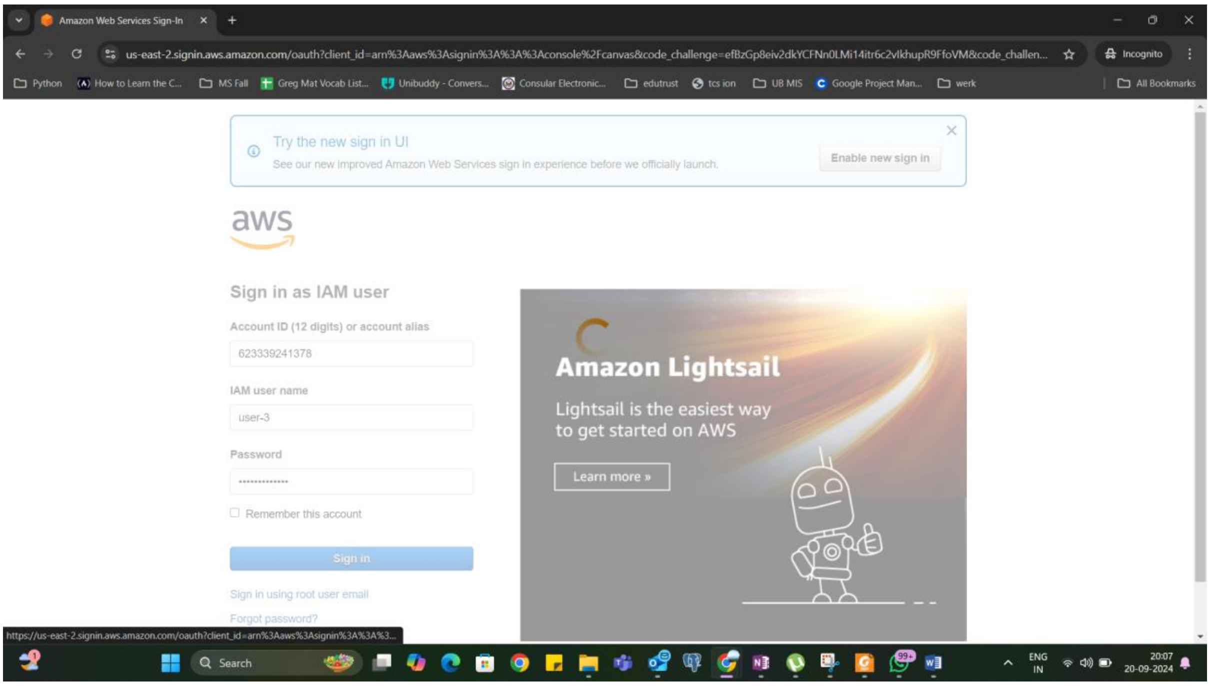

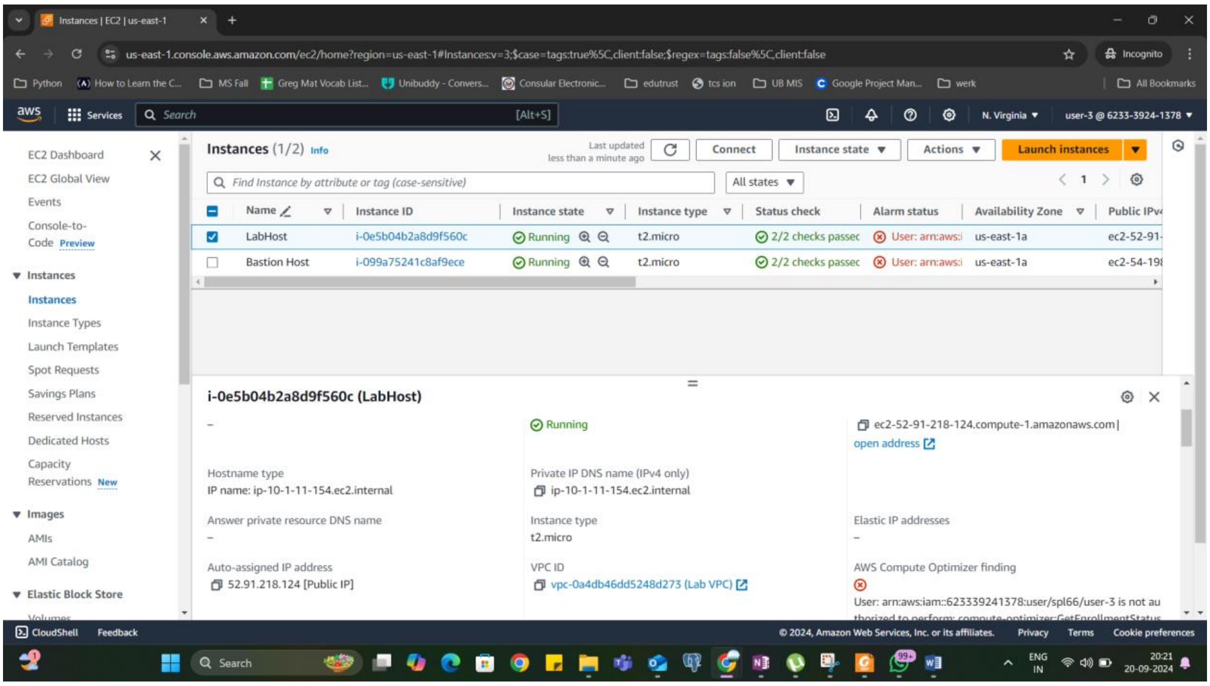
Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:v=3;$case=tags:true%5C,client:false;$regex=tags:false%5C,client:false
Instances (1/2) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv
LabHost
i-0e5b04b2a8d9f560c
Running
t2.micro
2/2 checks passed
us-east-1a
Bastion Host
i-099a75241c8af9ece
Running
t2.micro
2/2 checks passed
us-east-1a
i-0e5b04b2a8d9f560c (LabHost)
ec2-52-91-218-124.compute-1.amazonaws.com
open address
Hostname type
IP name: ip-10-1-11-154.ec2.internal
Private IP DNS name (IPv4 only)
ip-10-1-11-154.ec2.internal
Answer private resource DNS name
Instance type
t2.micro
Elastic IP addresses
Auto-assigned IP address
52.91.218.124 [Public IP]
VPC ID
vpc-0a4db46dd5248d273 (Lab VPC)
AWS Compute Optimizer finding
User: arn:aws:iam::623339241378:user/spl66/user-3 is not au
EC2 Dashboard
EC2 Global View
Events
Console-to-Code Preview
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations New
Images
AMIs
AMI Catalog
Elastic Block Store
CloudShell
Feedback
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Cookie preferences
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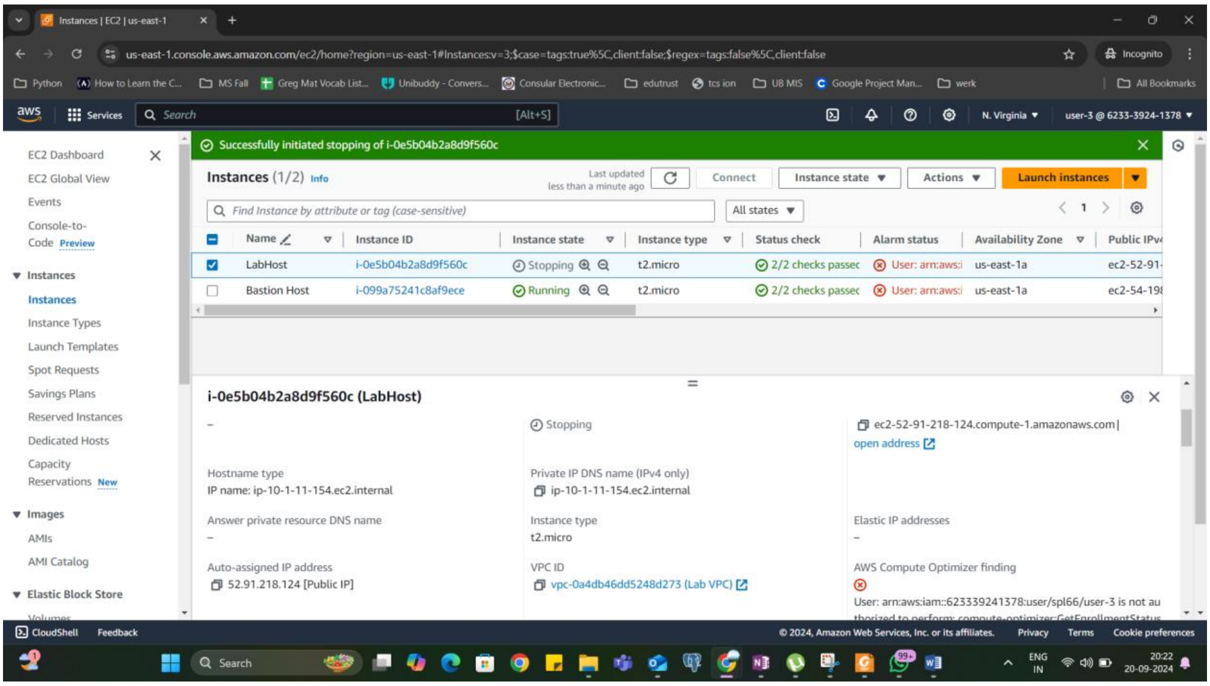
Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:v=3;$case=tags:true%5C,client:false;$regex=tags:false%5C,client:false
Successfully initiated stopping of i-0e5b04b2a8d9f560c
Instances (1/2) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv
LabHost
i-0e5b04b2a8d9f560c
Stopping
t2.micro
2/2 checks passed
us-east-1a
Bastion Host
i-099a75241c8af9ece
Running
t2.micro
2/2 checks passed
us-east-1a
i-0e5b04b2a8d9f560c (LabHost)
Stopping
ec2-52-91-218-124.compute-1.amazonaws.com
open address
Hostname type
IP name: ip-10-1-11-154.ec2.internal
Private IP DNS name (IPv4 only)
ip-10-1-11-154.ec2.internal
Answer private resource DNS name
Instance type
t2.micro
Elastic IP addresses
Auto-assigned IP address
52.91.218.124 [Public IP]
VPC ID
vpc-0a4db46dd5248d273 (Lab VPC)
AWS Compute Optimizer finding
User: arn:aws:iam::623339241378:user/spl66/user-3 is not au
CloudShell
Feedback
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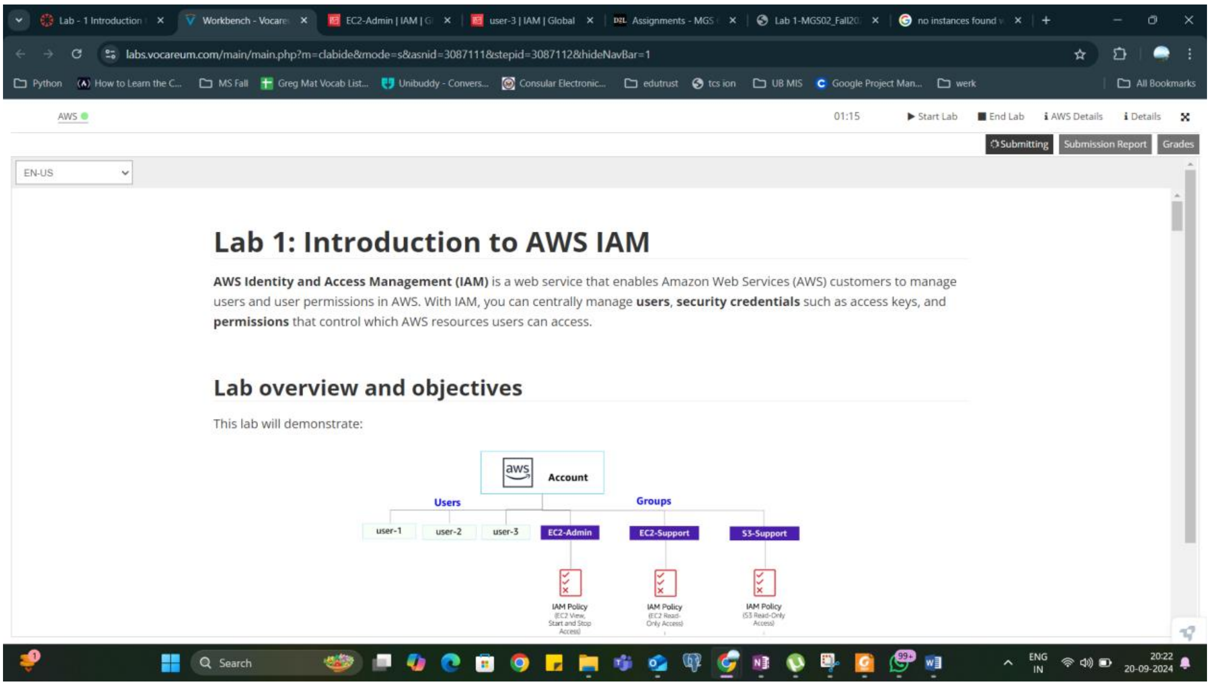

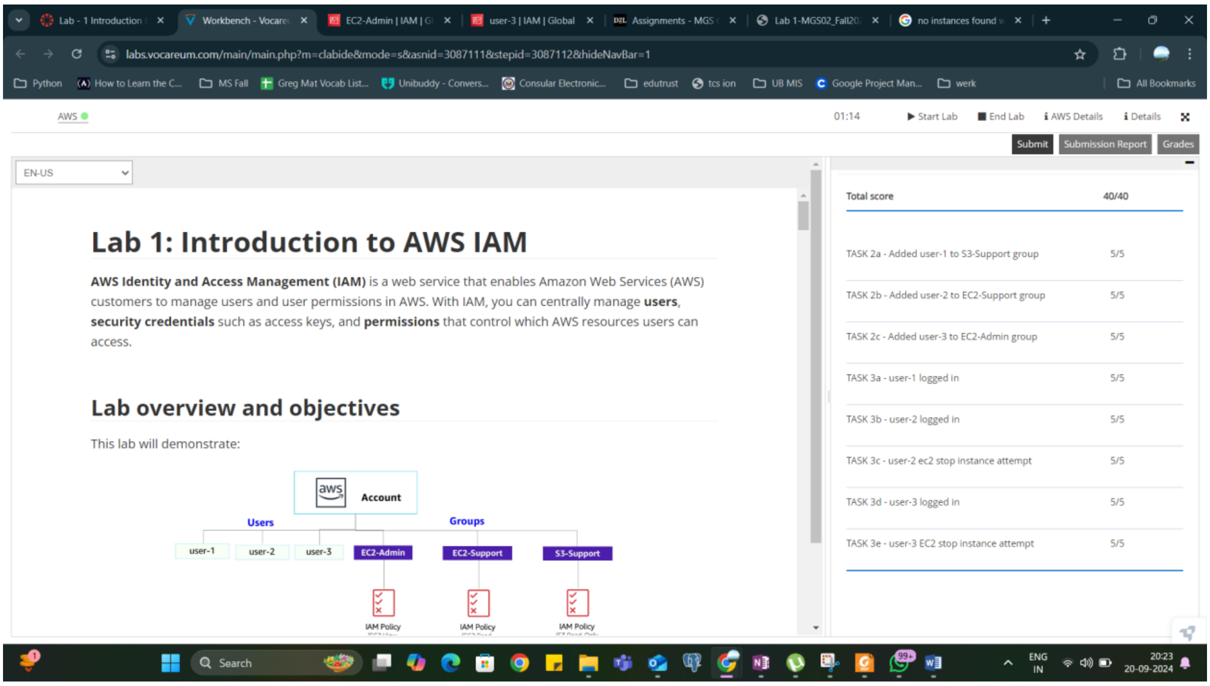

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